Insertion Sort Algorithm

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Introduction to Insertion Sort

Definition

Insertion sort is a simple sorting algorithm that builds the final sorted array one item at a time by comparisons.

How Insertion Sort Works

1. Start with the first element, assuming it is sorted.
2. Take the next element and compare it with elements in the sorted part.
3. Shift all the elements in the sorted part that are greater than the new element to the right.
4. Insert the new element into its correct position.
5. Repeat until all elements are sorted.

Pseudocode

```
Void insertion(int A[],n)

{ for(i=1;i<n;i++)

  {  temp = A[i];
     int j = i-1;

While(j>=0 && A[i]>temp)

    {  A[j+1] = A[i];
        j-- ;
      }

    A[j+1] = temp;
}
```

Pseudocode of insertion sort

Example

Step-1

43	31	26	29	12

Step-2

31	43	26	29	12

Step-3

26	31	43	29	12

Step-4

26	29	31	43	12

Step-5

12	26	29	31	43

Time Complexity of Insertion Sort

O(n^2)

Worst Case

When the array is reverse sorted.

O(n)

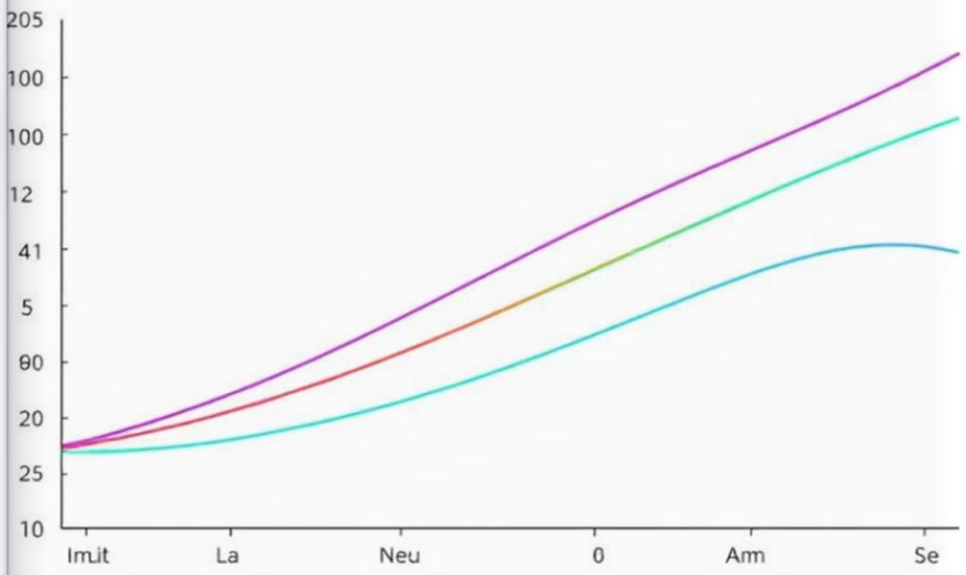
Best Case

When the array is already sorted.

O(n^2)

Average Case

The algorithm performs similarly to the worst case.



Advantages & Disadvantages of Insertion Sort

Simple to implement

Efficient for small datasets

Stable sort

Inefficient for large datasets

Higher time complexity

Conclusion

- Insertion sort is a basic sorting algorithm suitable for small or nearly sorted datasets.
- Understanding insertion sort helps in learning more complex algorithms.

O(n^2)

THANK YOU